

PRAJWAL MOHAN

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PROFESSIONAL SUMMARY

Quality and Validation Engineer with an M.S. in Biomedical Engineering (Arizona State University, 2025) and hands-on experience authoring Design History Files, executing verification and validation protocols, and managing nonconformance through formal change control. Applied ISO 13485, ISO 14971:2019, IEC 62304, and 21 CFR Part 820 across a full medical device development lifecycle as the sole Quality & Validation Engineer on an active device program. Targeting Quality Engineer, Validation Engineer, and Software Quality Engineer roles in medical device and biopharma.

TECHNICAL SKILLS

Quality & Regulatory: ISO 13485:2016 • ISO 14971:2019 • IEC 62304 • 21 CFR Part 820 • 21 CFR Part 803 • 21 CFR Part 11 • Design History File (DHF) • FMEA • CAPA • NCR / DCR • Root Cause Analysis • Design Controls • Change Control • GDP • Audit Readiness

Verification & Validation: V&V Planning (VVP) • Test Protocol Development (FTP / FTR) • Requirements Traceability (RTM) • Functional Verification Testing • IQ / OQ / PQ Concepts • Statistical Analysis • MATLAB • Minitab

Engineering & Technical: Python • Next.js / TypeScript / PostgreSQL • Prisma ORM • Google Gemini AI • EEG / Biosignal Processing • Wearable Medical Devices • Real-Time Data Streaming (LSL) • Git / GitHub

EDUCATION

Arizona State University, Tempe, AZ

December 2025

Master of Science, Biomedical Engineering • GPA: 3.87

RV College of Engineering, Bengaluru, India

May 2022

Bachelor of Engineering, Biotechnology • Wet lab: ELISA, HPLC, cell culture, PCR, fermentation

PROFESSIONAL EXPERIENCE

Quality and Validation Engineer

January 2026 – Present

adidas–ASU Center for Engagement Science, Arizona State University

- Served as sole Quality & Validation Engineer on an active closed-loop neurohaptic biofeedback device program (EATS) for deaf users, owning all quality documentation and verification activities from initiation through test closure.
- Authored a 7-document Design History File including Verification & Validation Plan (VVP), Software Requirements Specification (SRS), Functional Test Protocol (FTP), Functional Test Report (FTR), FMEA, Requirements Traceability Matrix (RTM), and Design Change Package aligned to ISO 13485:2016, ISO 14971:2019, IEC 62304, and 21 CFR Part 820.
- Defined system-level requirements across 5 hardware and software subsystems; achieved full bidirectional traceability from requirements through test results via RTM, ensuring every requirement was tested and every test was linked to a requirement.
- Identified a functional test failure during verification; raised a formal Nonconformance Report (NCR) with documented root cause analysis, then authored a Design Change Request (DCR) with full safety rationale and impact assessment. Drove the NCR through resolution: obtained PI approval, oversaw corrective design change, and re-executed testing to formal closure.
- Led a 10-failure-mode FMEA per ISO 14971:2019 risk management framework; updated all risk documentation following the design change to maintain post-change traceability.
- Supported an IRB-approved human-subject study protocol with zero adverse events reported.

Health Systems Strategy Analyst

January 2025 – December 2025

ASU Health, Arizona State University

- Supported development and coordination of large-scale healthcare funding proposals valued between \$20M–\$500M, including narrative development and compliance alignment.
- Partnered with ASU's Dean of Technology to develop an ethical AI governance framework addressing algorithmic safety, health equity, and responsible AI deployment in clinical environments.
- Analyzed federal healthcare policies and produced executive summaries supporting institutional decision-making across research and clinical leadership.
- Coordinated operational workflows for School of Medicine and Advanced Medical Engineering Grand Rounds programming.

PROJECTS

TraceFlow AI : AI-Enabled eQMS Platform

January 2025 – Present

- Independently designed and built a full-stack electronic Quality Management System (eQMS) from scratch, implementing five regulatory modules: Design Input & Risk Engine, Traceability Matrix / RTM Gap Analysis, CAPA Workflow, Post-Market Surveillance & Complaints, and Audit Trail with Electronic Signatures (21 CFR Part 11 compliant).
- Mapped all backend API routes, Prisma database schema, validation logic, and frontend pages to ISO 13485:2016, ISO 14971:2019, IEC 62304, 21 CFR Part 820, 21 CFR Part 803, and 21 CFR Part 11 enforcing regulatory requirements as system behavior, not optional fields.
- Completed a full regulatory overhaul rewriting all five modules to align with FDA and international standards; demonstrates end-to-end understanding of QMS architecture as a working software system rather than documentation.

C.E.R.A.- Clinical Equipment Retrieval Assistant

Sep 2025 – Present

- Developed an enterprise-grade Retrieval-Augmented Generation (RAG) system to support biomedical and clinical engineers in troubleshooting ICU ventilators using manufacturer service manuals; built hybrid semantic + keyword retrieval pipelines enabling precise identification of error codes, part numbers, and repair procedures.
- Implemented risk-management guardrails inspired by FDA and ISO 14971 principles automated extraction and prioritization of WARNING / CAUTION / DANGER safety statements, device-aware cross-instruction validation logic to prevent errors, and source traceability citations in every response.
- Engineered automated generation of CMMS-ready maintenance work orders and structured diagnostic checklists, converting unstructured technical documentation into actionable clinical engineering workflows.

Emotion-Aware Tactile Feedback System (EATS)

May 2025 – Dec 2025

- Designed and built a closed-loop wearable biofeedback system integrating real-time EEG biosignal acquisition with adaptive haptic feedback to enhance emotional perception for deaf users during live acoustic events.
- Developed real-time signal processing pipelines computing Frontal Alpha Asymmetry (FAA) and Theta/Beta Ratio (TBR) to dynamically modulate tactile stimulation intensity based on detected emotional state.
- Implemented experimental evaluation workflows and analyzed physiological and interaction data to assess system responsiveness; iteratively refined feedback mapping algorithms based on user engagement outcomes.

CAPTAVATE — Wearable EEG Attention-Detection Device

Jan 2024 – May 2024

- Executed design verification testing on attention-detection classification algorithms by correlating predicted vs. observed outputs across controlled stimulus conditions to validate neurofeedback accuracy.
- Built validation datasets and performed statistical performance analysis in MATLAB; authored technical validation reports and supporting documentation following Good Documentation Practices (GDP) and device development lifecycle requirements.

CERTIFICATIONS & TRAINING

- Quality Management for Operational Excellence — LinkedIn Learning, June 2026
- Leading Quality in Patient Safety: Systems Thinking and Design — LinkedIn Learning, June 2026
- Six Sigma Yellow Belt — LinkedIn Learning, June 2026
- Learning Minitab — LinkedIn Learning, August 2025
- Quality Management Foundations — LinkedIn Learning, August 2025